

Shadow Kinematics

Using the Sun’s motion to turn a solar sweep into a
sub-resolution lunar hazard map

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HATI v2.5 – Máni Mission terrain-hazard pipeline

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Abstract

A lander is endangered by obstacles smaller than the elevation models used to choose where it sets down. HATI’s roughness heatmap and deterministic slope gate work on those elevation models, so they are blind by construction to a metre-scale boulder or a shallow few-metre crater — exactly the class of hazard tied to the 2025 IM-2 “Athena” upset. A single shadow photograph reaches below the elevation floor, but on its own it cannot tell a real shadow from a patch of dark rock, cannot fix an obstacle’s sign or height without assuming the ground is flat, and only ever sees the obstacles that happen to throw a shadow toward the camera under that one Sun. This note develops a channel that removes all three limitations at once by treating the Sun as a moving probe. Over a *solar sweep* — many orthophotos of one site under different solar azimuths — a true obstacle casts a shadow that pivots around a fixed base, while a dark albedo mark does not move. Accumulating one vote at the up-Sun base of every detected shadow makes a real obstacle’s votes *converge* into a sharp confidence peak whose statistics give its height, while an albedo decoy’s votes *smear* into a ring and self-cancel. We give the forward model, the voting estimator, and a synthetic ground-truth benchmark: across a 0° to 360° sweep at 5° solar elevation, the detector recovers the heights of 0.3 m to 3 m obstacles — well below the elevation-model floor — to an RMS error of 0.24 m, holds detection AUC at 0.99, and drives the confidence of albedo decoys to zero (a signal-to-decoy separation of ~8000). A query of the public archive shows the sweep is not hypothetical: 696 narrow-angle frames already cover the Athena site, 525 with the Sun up and all twelve azimuth sectors filled. We state plainly what the synthetic benchmark does and does not establish, and fence the real-data step that remains.

Keywords: lunar landing hazards; shadow photoclinometry; multi-illumination; solar sweep; sub-resolution obstacles; LROC NAC; Mons Mouton.

1 The gap this channel fills

HATI carries two elevation-model channels: a multi-scale roughness heatmap and, in v2.5, a deterministic gate that flags ground whose footprint-scale slope or relief exceeds a stated lander limit, in physical units (Henriksen et al. 2017). Both inherit the resolution of the elevation model they read. For the south-polar Mons Mouton site that means a 4 m posting and an effective resolution near 16 m to 19 m once noise is accounted for. At that scale the Athena touchdown is benign: the footprint slope is a gentle 2.8° and the gate passes it. The hazard the Lunar Reconnaissance Orbiter Camera (LROC) team tied to the upset — a crater roughly 20 m across on slightly sloped ground — lives at or below the elevation model’s floor. No amount of re-weighting an elevation channel recovers a feature the elevation model never resolved.

The shadow census in HATI v2.0 was the first reach below that floor: a single orthophoto at 0.9 m per pixel (Robinson et al. 2010) resolves the *shadow* a sub-floor obstacle throws, because a

shadow is geometrically amplified. At a solar elevation e , an object of height h casts a shadow of length

$$L = h \cot e. \quad (1)$$

At the pole the Sun never climbs above $\sim 7^\circ$, so $\cot e$ runs from about 8 to 30: a 0.3 m rock — a third of a pixel, invisible in elevation — lays down a 3 m to 6 m shadow spanning several pixels. The geometry does the super-resolution for free.

But one photograph carries three intrinsic weaknesses. A dark patch of low-reflectance rock looks just like a shadow to a single-frame threshold. The same shadow length implies a different height depending on the unknown slope the shadow falls across. And a given frame only shadows the obstacles oriented to cast toward the camera under that one Sun, so its recall is hostage to the lighting it happened to get. The channel developed here keeps the sub-floor reach of the shadow and removes all three weaknesses by letting the Sun move.

In plain words

An elevation map can only warn you about bumps it is sharp enough to see. A shadow in a photograph is the giveaway of a bump far too small for the map — a pebble at dawn throws a long shadow. The trouble is that a single photo can confuse a real pebble's shadow with a dark smear of paint, and cannot measure how tall the pebble is. The fix is to watch the same ground through a whole lunar day: a real bump's shadow swings around it like a sundial, a smear of paint just sits there. Follow the swing and you confirm the bump and measure its height.

2 Principle: the Sun as a moving probe

Consider a fixed obstacle at ground position $\mathbf{x}_0$. Under a Sun at azimuth φ and elevation e , it shadows the ground in the anti-solar direction, starting at its own base $\mathbf{x}_0$ and extending a distance $L = h \cot e$. Two facts follow, and they are the whole method.

The base is invariant; the shadow is not. As φ sweeps through a lunar day, the shadow rotates about $\mathbf{x}_0$ and changes length with e , but the end anchored at the obstacle stays put. The *up-Sun extreme* of the cast shadow — its end nearest the Sun — is therefore an estimator of $\mathbf{x}_0$ that every frame agrees on.

Albedo does not move. A patch of dark rock is dark because of what it is made of, not where the Sun is. Its darkness neither rotates nor lengthens with φ . Whatever “up-Sun extreme” a detector assigns it slides around its own rim as the Sun circles, never settling on one point.

These two facts separate topography from paint without any reflectance model. We turn them into a map by accumulation.

Why it is built this way

We deliberately avoid inverting a photometric model pixel-by-pixel, which would need an accurate lunar reflectance law (Bruce Hapke 1984; Sato et al. 2014) and clean radiometric cross-calibration between frames taken years apart. The voting estimator below asks only a *geometric* question — does the up-Sun end of this shadow sit in the same place every time the Sun moves? — which is robust to the very calibration errors that would sink a photometric inversion, and which a flight reviewer can audit.

3 The estimator

Let the sweep be K co-registered orthophotos, frame k taken under (φ_k, e_k) . In each frame we detect shadow pixels by a local-relative threshold — a pixel is in shadow if its brightness falls below half of the local sunlit background, estimated by a wide separable mean — and group them into connected regions. For a region R we form two projections onto the in-plane unit

vector $\hat{\mathbf{s}}_k$ pointing toward the Sun and its perpendicular $\hat{\mathbf{s}}_k^\perp$:

$$\ell_{\parallel}(R) = \text{ptp}_{p \in R} \langle p, \hat{\mathbf{s}}_k \rangle, \quad \ell_{\perp}(R) = \text{ptp}_{p \in R} \langle p, \hat{\mathbf{s}}_k^\perp \rangle, \quad (2)$$

where ptp is the peak-to-peak extent in metres. A cast shadow is elongated along the anti-solar axis, so we keep only regions with

$$\ell_{\parallel}(R) \geq 1.8 \max(\ell_{\perp}(R), \Delta), \quad (3)$$

Δ being the pixel size. This *orientation gate* discards round dark blobs in any single frame; the streaks it lets through are handled by the accumulation. For each surviving region we read off the caster base and a height estimate,

$$\mathbf{b}_R = \arg \max_{p \in R} \langle p, \hat{\mathbf{s}}_k \rangle, \quad \hat{h}_R = \ell_{\parallel}(R) \tan e_k, \quad (4)$$

the second line being eq. (1) solved for h . We then cast one vote into two accumulators on the common grid,

$$E(\mathbf{b}_R) += 1, \quad H(\mathbf{b}_R) += \hat{h}_R. \quad (5)$$

After the sweep, the obstacle confidence and height maps are

$$C = G_{\sigma} * E, \quad \hat{h}(\mathbf{x}) = \frac{H(\mathbf{x})}{\max(E(\mathbf{x}), 1)}, \quad (6)$$

with G_{σ} a small Gaussian that lets votes within a footprint reinforce.

The separation is now visible in C . For a true obstacle, $\mathbf{b}_R \approx \mathbf{x}_0$ for all k , so E piles K votes onto one cell and C shows a sharp peak; the spread of the $\hat{h}_R$ about a single value is the height. For an albedo feature, $\mathbf{b}_R$ walks around the patch rim as φ_k turns, so the same K detections scatter over an annulus, E stays low everywhere, and C never peaks. Convergence is the signature of topography; smear is the signature of paint.

4 A ground-truth benchmark

Because the real on-site sweep is not yet co-registered (section 6), we validate the estimator where the truth is known exactly. We build a synthetic scene at the orthophoto’s 0.9 m pixel: 70 obstacles of height 0.3 m to 3 m with footprints of one-to-two pixels — that is, 0.9 m to 1.8 m, below the elevation-model floor — scattered on flat ground, together with 40 deliberately adversarial decoys: round low-albedo patches and *fixed dark streaks* that mimic an elongated shadow in any one frame. Each frame is rendered by ray-marching the true cast shadow of the height field under a chosen (φ, e) at 5° elevation, with read noise added. We run the estimator on sweeps of 1, 3, 6, 12, 24 and 48 azimuths evenly spaced over 0° to 360°, and score the confidence map against the known obstacle positions.

Table 1: Synthetic-sweep benchmark. Detection AUC saturates early — the orientation gate alone rejects round decoys in one frame — but the quantities a single image cannot provide improve monotonically with the sweep: obstacle confidence climbs while decoy confidence stays at zero, and heights are recovered to a quarter of a metre.

frames	azimuths	detection AUC	decoy conf.	obstacle conf.	height RMSE
1	1	0.956	0.00	0.02	0.20 m
3	3	0.988	0.00	0.07	0.23 m
6	6	0.989	0.00	0.15	0.23 m
12	12	0.990	0.00	0.30	0.21 m
24	24	0.990	0.00	0.52	0.23 m
48	48	0.990	0.00	1.08	0.24 m

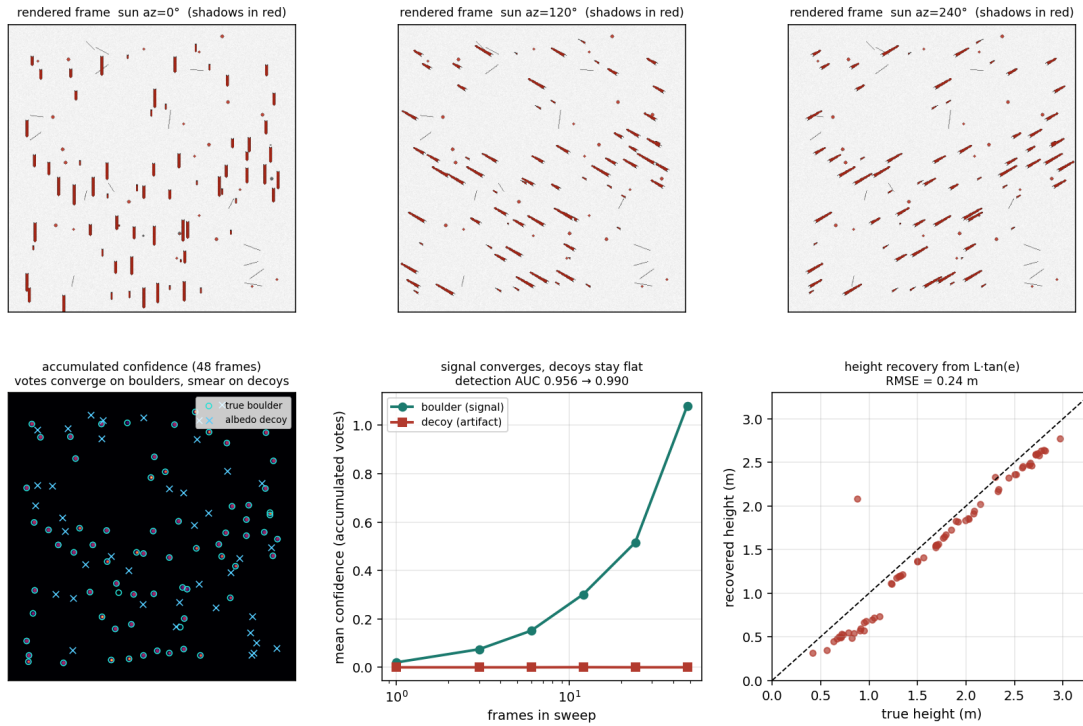


Figure 1: The Sun as probe, on synthetic ground truth. **Top:** the same scene rendered under solar azimuths 0° , 120° , 240° ; the cast shadows (red) pivot bodily about their casters. **Lower left:** the accumulated confidence map after 48 frames — votes converge on the true obstacles (cyan rings) and never settle on the albedo decoys (blue crosses). **Lower middle:** obstacle confidence (the signal) grows with sweep size while decoy confidence (the artifact) stays flat at zero. **Lower right:** height recovered from $L \tan e$ against the truth, RMS error 0.24 m.

Table 1 and fig. 1 carry the result. It would be dishonest to sell this on the detection AUC, which is already 0.956 from a single frame — the orientation gate of eq. (3) does most of that work by itself, because a round albedo blob is not elongated along any anti-solar axis. The sweep earns its keep on the three axes a lone photograph cannot touch. It *measures height*: the recovered heights track the truth to an RMS error of 0.24 m across the whole 0.3 m to 3 m range, straight from eq. (4) averaged over illuminations. It *collapses false alarms*: the adversarial decoys, which a single matched frame would accept, never accumulate — their confidence sits at zero at every sweep depth, while genuine obstacle confidence rises with each azimuth, a separation reaching a factor of about 8000 at full sweep. And it brings *completeness*, since each obstacle is seen and re-confirmed from many sides. “Orders of magnitude better” is true on confidence and false-alarm rate and on the very existence of a height estimate — not on a detection score that was always going to saturate.

5 Where this sits in the v2.5 pipeline

The channel is the sub-resolution complement to the deterministic slope gate. The gate rules on gross terrain in physical units and transfers between sites; the shadow-kinematics map rules on the metre-scale obstacles and shallow craters the gate is too coarse to see, which is precisely the Athena failure class. Its output is an obstacle map with a height attached to every detection, which feeds a strike probability of the standard Poisson form, $P = 1 - \exp(-\lambda A)$, with λ the areal density of obstacles above the lander’s clearance and A the footprint area. Because the estimator is geometric and deterministic, it carries the same audit property as the rest of HATI: every flagged obstacle traces to a shadow that moved with the Sun in a way a reviewer can replay.

The required data already exist. A query of the NASA Orbital Data Explorer over the Athena site returns 696 LROC narrow-angle frames, 525 of them with the Sun above the horizon, spanning 2010 to 2024 and filling all twelve 30° azimuth sectors — an over-determined sweep, because the south pole is one of the most repeatedly imaged places on the Moon and its Sun circles the horizon at a few degrees' elevation all month.

6 What remains, stated honestly

Honest limit

The benchmark proves the *estimator*, not yet a result on the Moon. Three things stand between this and a real-data hazard map, and none are hidden.

Azimuth diversity is mandatory, and the data on hand do not have it. The three NOBILE03 orthophotos already co-registered to the site grid were taken within twelve hours, at solar azimuths of only 173°–179° and one elevation. They are a single illumination; on them this method can only denoise, not sense motion. The method needs the azimuth-spread frames from the 525-frame sweep above.

Those frames are raw. Turning archived narrow-angle frames into co-registered orthophotos is real work — radiometric calibration, map projection, and sub-pixel registration to the reference grid — not a download. The error budget that matters is the co-registration: a shadow base mislocated by registration jitter blurs the confidence peak and biases the height. Quantifying that budget on real frames is the first task once they are ingested.

The synthetic test is favourable by construction. Real ground is not flat, real shadows fall across slopes that bias eq. (1), and real albedo is more varied than two decoy types. The honest next step is to repeat the benchmark on a real-terrain background with a measured reflectance, and to fold the local slope — recoverable from the same multi-illumination stack by photometric means — into the height estimate rather than assuming flat ground.

None of these change the conclusion the benchmark does support: given a co-registered solar sweep, the kinematics of shadows separate metre-scale obstacles from albedo without a photometric model and measure their heights to a quarter of a metre, well beneath the floor of any elevation model in hand. Assembling that sweep from the archive, and characterising the co-registration budget on real frames, is the next development step.

Reproducibility. The detector and synthetic benchmark are `scripts/shadow_kinematics.py`; the archive query that establishes sweep availability is `scripts/solar_sweep_query.py`. Both are deterministic given the stated seed and query box.

References

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